

BTEC Level 3 Applied Science · Unit 1 Chemistry · Answer sheet

Ionic Bonding — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

A–E

1. Electrostatic attraction between oppositely charged ions.
2. A metal and a non-metal.
3. Ions are not free to move.
4. Na loses $1\text{ e}^- \rightarrow \text{Na}^+$. Cl gains that electron $\rightarrow \text{Cl}^-$. Oppositely charged ions attract.
5. Strong attractions throughout a giant lattice need a lot of energy to overcome.
6. Ions become free to move and carry charge.
7. Na_2O ; MgCl_2 ; Al_2O_3 .
8. A shift lines up like charges, which repel, so the crystal splits.
9. Molten NaCl: mobile ions. Copper: delocalised electrons.
10. Mg loses $2\text{e}^- \rightarrow \text{Mg}^{2+}$. O gains $2\text{e}^- \rightarrow \text{O}^{2-}$. Electrostatic attraction. Giant lattice. [4]
11. Strong ionic attractions $\rightarrow$ high m.p. Solid ions fixed $\rightarrow$ no conduction. [4]
12. $\text{Al}_2(\text{SO}_4)_3$ so +6 balances -6. [2]
13. Mg^{2+} and O^{2-} have higher charges than Na^+ and Cl^- , so attraction is stronger.